

CHAPTER 2 I

portfolio management



The aim of this Chapter...

...is to move beyond the world of risk-free hedging and enter the exciting and dangerous world of gambling, also known as investing, in risky assets. You will see some simple ideas for deciding how to allocate your money between all the possible investments on offer.

In this Chapter...

- Modern Portfolio Theory and the Capital Asset Pricing Model
- optimizing your portfolio
- alternative methodologies such as cointegration
- how to analyze portfolio performance

21.1 INTRODUCTION

The theory of derivative pricing is a theory of deterministic returns: we hedge our derivative with the underlying to eliminate risk, and our resulting risk-free portfolio then earns the risk-free rate of interest. Banks make money from this hedging process; they sell something for a bit more than it's worth and hedge away the risk to make a guaranteed profit.

But not everyone is hedging. Fund managers buy and sell assets (including derivatives) with the aim of beating the bank's rate of return. In so doing they take risk. In this chapter I explain some of the theories behind the risk and reward of investment. Along the way I show the benefits of diversification, how the return and risk on a portfolio of assets is related to the return and risk on the individual assets, and how to optimize a portfolio to get the best value for money.

For the most part, the assumptions are as follows:

- We hold a portfolio for 'a single period,' examining the behavior after this time.
- During this period returns on assets are Normally distributed.
- The return on assets can be measured by an expected return (the drift) for each asset, a standard deviation of return (the volatility) for each asset and correlations between the asset returns.

21.2 DIVERSIFICATION

In this section I introduce some more notation, and show the effects of diversification on the return of the portfolio.

We hold a portfolio of N assets. The value today of the i th asset is S_i and its random return is R_i over our time horizon T . The R s are Normally distributed with mean $\mu_i T$ and standard deviation $\sigma_i \sqrt{T}$. The correlation between the returns on the i th and j th assets is ρ_{ij} (with $\rho_{ii} = 1$). The parameters μ , σ and ρ correspond to the drift, volatility and correlation that we are used to. Note the scaling with the time horizon.



If we hold w_i of the i th asset, then our portfolio has value

$$\Pi = \sum_{i=1}^N w_i S_i.$$

At the end of our time horizon the value is

$$\Pi + \delta \Pi = \sum_{i=1}^N w_i S_i (1 + R_i).$$

We can write the relative change in portfolio value as

$$\frac{\delta \Pi}{\Pi} = \sum_{i=1}^N w_i R_i, \quad (21.1)$$

where

$$W_i = \frac{w_i S_i}{\sum_{i=1}^N w_i S_i}.$$

The weights W_i sum to one.

From (21.1) it is simple to calculate the expected return on the portfolio

$$\mu_{\Pi} = \frac{1}{T} E \left[\frac{\delta \Pi}{\Pi} \right] = \sum_{i=1}^N W_i \mu_i \quad (21.2)$$

and the standard deviation of the return

$$\sigma_{\Pi} = \frac{1}{\sqrt{T}} \sqrt{\text{var} \left[\frac{\delta \Pi}{\Pi} \right]} = \sqrt{\sum_{i=1}^N \sum_{j=1}^N W_i W_j \rho_{ij} \sigma_i \sigma_j}. \quad (21.3)$$

In these, we have related the parameters for the individual assets to the expected return and the standard deviation of the entire portfolio.

21.2.1 Uncorrelated assets

Suppose that we have assets in our portfolio that are uncorrelated, $\rho_{ij} = 0$, $i \neq j$. To make things simple assume that they are equally weighted so that $W_i = 1/N$. The expected return on the portfolio is represented by

$$\mu_{\Pi} = \frac{1}{N} \sum_{i=1}^N \mu_i,$$

the average of the expected returns on all the assets and the volatility becomes

$$\sigma_{\Pi} = \sqrt{\frac{1}{N^2} \sum_{i=1}^N \sigma_i^2}.$$

This volatility is $O(N^{-1/2})$ since there are N terms in the sum. As we increase the number of assets in the portfolio, the standard deviation of the returns tends to zero. It is rather extreme to assume that all assets are uncorrelated but we will see something similar when I describe the Capital Asset Pricing Model below, diversification reduces volatility without hurting expected return.

I am now going to refer to volatility or standard deviation as **risk**, a bad thing to be avoided (within reason), and the expected return as **reward**, a good thing that we want as much of as possible.



Time Out...

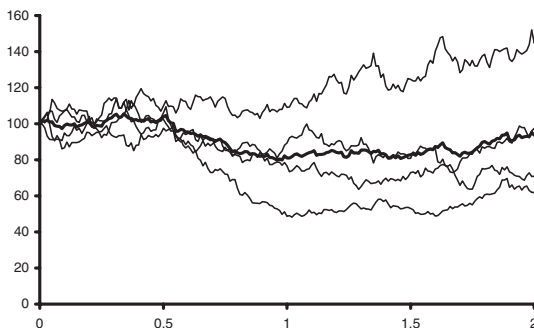
Spreadsheet test

In the following spreadsheet you can see the effect of investing all your cash in one risky asset, or of spreading it across four equally risky but uncorrelated assets. A convincing case for diversification... but watch out for those market crashes when all correlations become one.



Simple example of diversification

	A	B	C	D	E	F	G	H	I
1	Drift	0.1		Time	Asset 1	Asset 2	Asset 3	Asset 4	Basket
2	Volatility	0.25		0	100	100	100	100	100
3	Timestep	0.01		0.01	101.4946	102.0604	99.80029	102.4483	101.4509
4			=D2+\$B\$3	0.02	103.563	100.4693	97.82029	104.2275	101.52
5				0.03	105.7847	102.3088	93.20119	106.3945	101.9223
6				0.04	99.71384	107.1702	91.29877	106.0402	101.0557
7				0.05	92.86761	106.9663	91.14639	113.5807	101.1402
8			=E6*(1+\$B\$1*\$B\$3+\$B\$2*SQRT(\$B\$3	0.06	90.58581	102.5303	91.3095	111.6031	99.00717
9			)*(RAND()+RAND()+RAND()+RAND()+	0.07	93.42845	100.7858	91.47827	108.5943	98.57169
10			RAND()+RAND()+RAND()+RAND()+R	0.08	=AVERAGE(E8:H8)		91.14564	107.5179	98.5553
11			AND()+RAND()+RAND()+RAND()-6))	0.09	90.62362	105.6249	86.09578	106.91	97.31358
12								107.3784	98.98025
13								108.2679	99.85746
14								110.9582	99.38457
15								108.1393	99.22791
16								108.0873	100.0265
17								105.3345	98.15725
18								107.7298	98.97677
19								107.3138	99.37793
20								107.0215	100.223
21								107.0683	100.8625
22								101.59	100.637
23								100.8669	98.90859
24								99.30773	97.84018
25								101.2298	99.08378
26								103.2705	98.91596
27								104.8602	98.64734
28								111.2662	101.0386
29								112.5718	101.8489
30				0.28	105.3536	95.48459	96.4614	114.2369	102.8841
31				0.29	105.6262	93.65618	100.6464	112.8307	103.1899
32				0.3	110.304	92.95482	100.7716	114.2496	104.57
33				0.31	113.8359	92.39261	101.0787	114.2613	105.3921



21.3 MODERN PORTFOLIO THEORY

We can use the above framework to discuss the ‘best’ portfolio. The definition of ‘best’ was addressed very successfully by Nobel Laureate Harry Markowitz. His model provides a way of defining portfolios that are **efficient**. An efficient portfolio is one that has the highest reward for a given level of risk, or the lowest risk for a given reward. To see how this works imagine that there are four assets in the world, A, B, C and D, with reward and risk as shown in Figure 21.1 (ignore E for the moment). If you could buy any one of these (but as yet you are not allowed more than one), which would you buy? Would you choose D? No, because it has the same risk as B but less reward. It has the same reward as C but for a higher risk. We can rule out D. What about B or C? They are both appealing when set against D, but against each other it is not so clear. B has a higher risk, but gets a higher reward. However, comparing them both with A we see that there is no contest. A is the preferred choice. If we introduce asset E with the same risk as B and a higher reward than A, then we cannot objectively say which out of A and E is the better, this is a subjective choice and depends on an investor’s **risk preferences**.

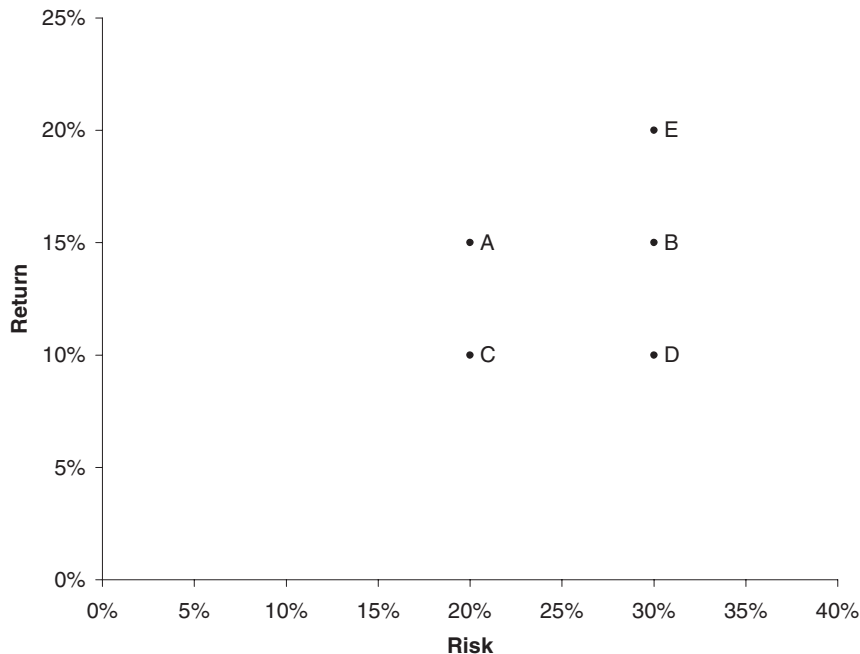


Figure 21.1 Risk and reward for five assets.

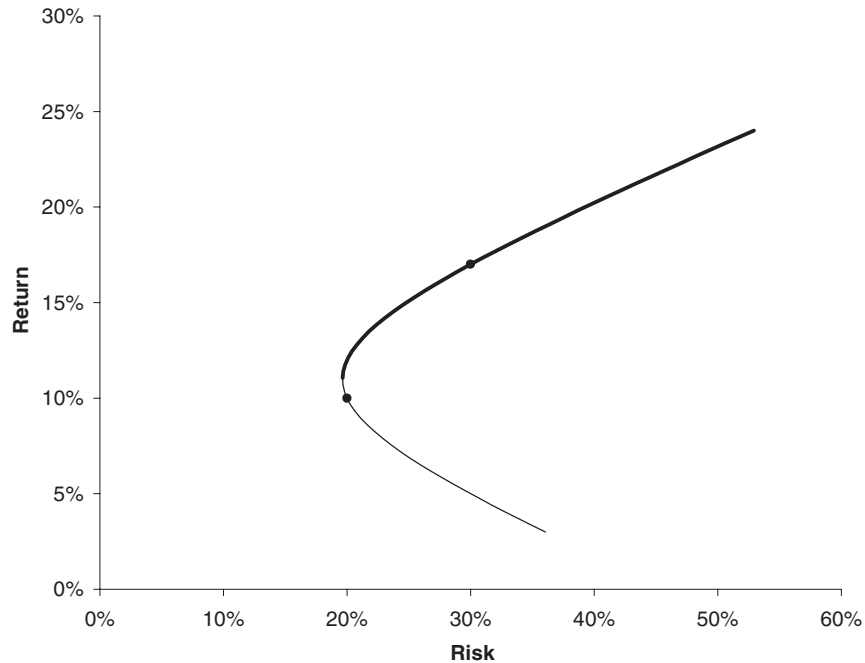


Figure 21.2 Two assets and any combination.

From (21.2) and (21.3) we have

$$\mu_{\Pi} = W\mu_A + (1 - W)\mu_E$$

and

$$\sigma_{\Pi}^2 = W^2\sigma_A^2 + 2W(1 - W)\rho\sigma_A\sigma_E + (1 - W)^2\sigma_E^2.$$

Here W is the weight of asset A, and, remembering that the weights must add up to one, the weight of asset E is $1 - W$.



Demonstration of optimal asset allocation

As we vary W , so the risk and the reward change. The line in risk/reward space that is parameterized by W is a hyperbola, as shown in Figure 21.2. The part of this curve in bold is efficient, and is preferable to the rest of the curve. Again, an individual's risk preferences will say where he wants to be on the bold curve. When one of the volatilities is zero the line becomes straight. Anywhere on the curve between the two dots requires a long position in each asset. Outside this region, one of the assets is sold short to finance the purchase of the other. Everything that follows assumes that we can sell short as much of an asset as we want. The results change slightly when there are restrictions.

If we have many assets in our portfolio we no longer have a simple hyperbola for our possible risk/reward profiles, instead we get something like that shown in Figure 21.3. This figure now uses *all* of A, B, C, D and E, not just the A and E. Even though B, C and D are not individually appealing

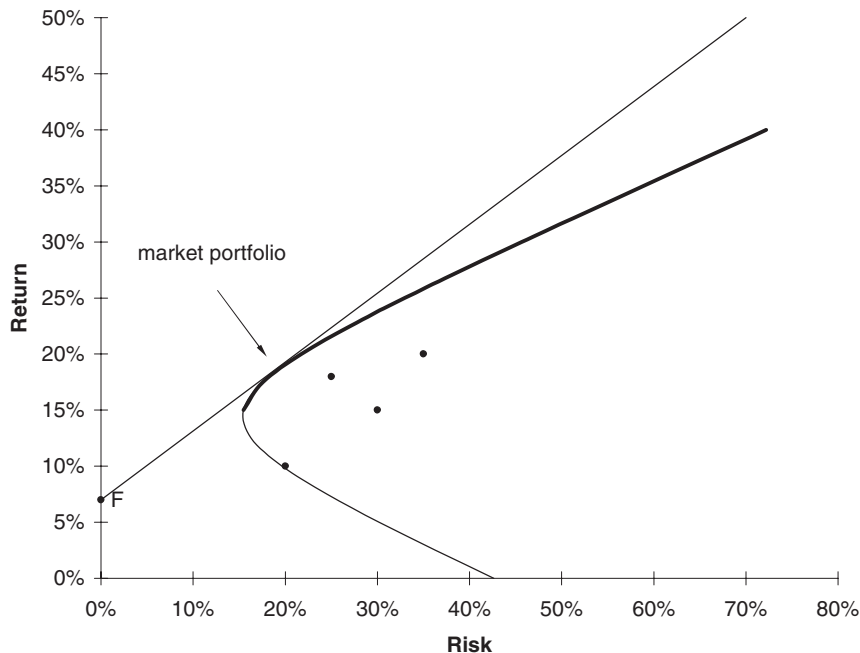


Figure 21.3 Portfolio possibilities and the efficient frontier.

they may well be useful in a portfolio, depending how they correlate, or not, with other investments. In this figure we can see the **efficient frontier** marked in bold. Given any choice of portfolio we would choose to hold one that lies on this efficient frontier.

The calculation of the risk for a given return is demonstrated in the spreadsheet in Figure 21.4. This spreadsheet can be used to find the efficient frontier if it is used many times for different target returns.

21.3.1 Including a risk-free investment

A risk-free investment earning a guaranteed rate of return r would be the point F in Figure 21.3. If we are allowed to hold this asset in our portfolio, then since the volatility of this asset is zero, we get the new efficient frontier which is the straight line in the figure. The portfolio for which the straight line touches the original efficient frontier is called the **market portfolio**. The straight line itself is called the **capital market line**.¹

21.4 WHERE DO I WANT TO BE ON THE EFFICIENT FRONTIER?

Having found the efficient frontier we want to know whereabouts on it we should be. This is a personal choice; the efficient frontier is objective, given the data, but the 'best' position on it is subjective.

¹ In the risk-neutral world they think that all investments lie on the horizontal line going through the point $(0, r)$.

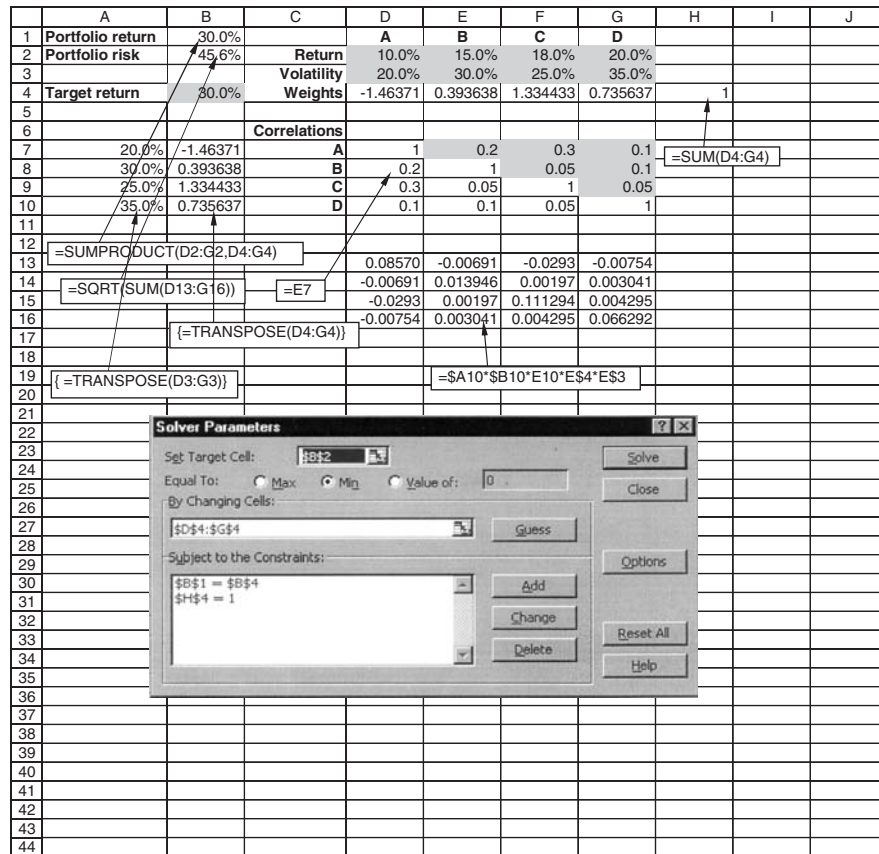


Figure 21.4 Spreadsheet for calculating one point on the efficient frontier.

The following is a way of interpreting the risk/reward diagram that may be useful in choosing the best portfolio.

The return on portfolio Π is Normally distributed because it is comprised of assets which are themselves Normally distributed. It has mean μ_{Π} and standard deviation σ_{Π} (I have ignored the dependence on the horizon T).

The slope of the line joining the portfolio Π to the risk-free asset is

$$s = \frac{\mu_{\Pi} - r}{\sigma_{\Pi}}.$$

This is an important quantity, it is a measure of the likelihood of Π having a return that exceeds r . If $C(\cdot)$ is the cumulative distribution function for the standardized Normal distribution then $C(s)$ is the probability that the return on Π is at least r . More generally

$$C\left(\frac{\mu_{\Pi} - r^*}{\sigma_{\Pi}}\right)$$

is the probability that the return exceeds r^* . This suggests that if we want to minimize the chance of a return of less than r^* we should choose the portfolio from the efficient frontier set Π_{eff} with the largest value of the slope

$$\frac{\mu_{\Pi_{\text{eff}}} - r^*}{\sigma_{\Pi_{\text{eff}}}}.$$

Conversely, if we keep the slope of this line fixed at s then we can say that with a confidence of $C(s)$ we will lose no more than

$$\mu_{\Pi_{\text{eff}}} - S\sigma_{\Pi_{\text{eff}}}.$$

Our portfolio choice could be determined by maximizing this quantity. These two strategies are shown schematically in Figure 21.5.

Neither of these methods give satisfactory results when there is a risk-free investment among the assets and there are unrestricted short sales, since they result in infinite borrowing.

Another way of choosing the optimal portfolio is with the aid of a **utility function**. This approach is popular with economists. In Figure 21.6 I show **indifference curves** and the

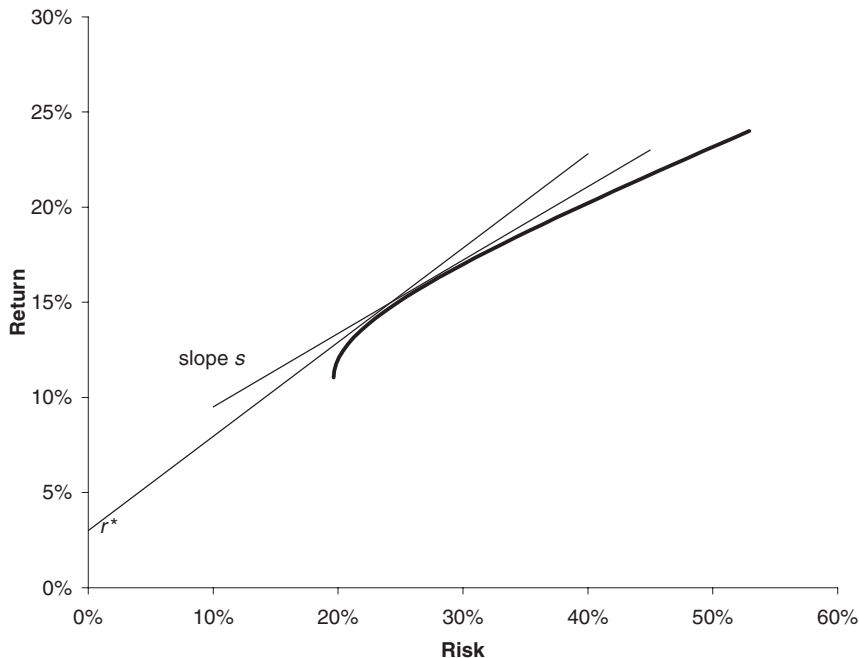
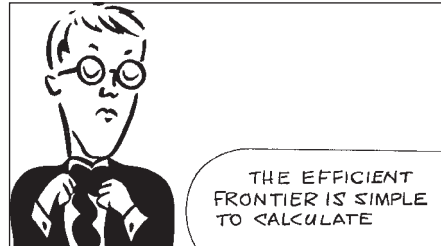


Figure 21.5 Two simple ways for choosing the best efficient portfolio.

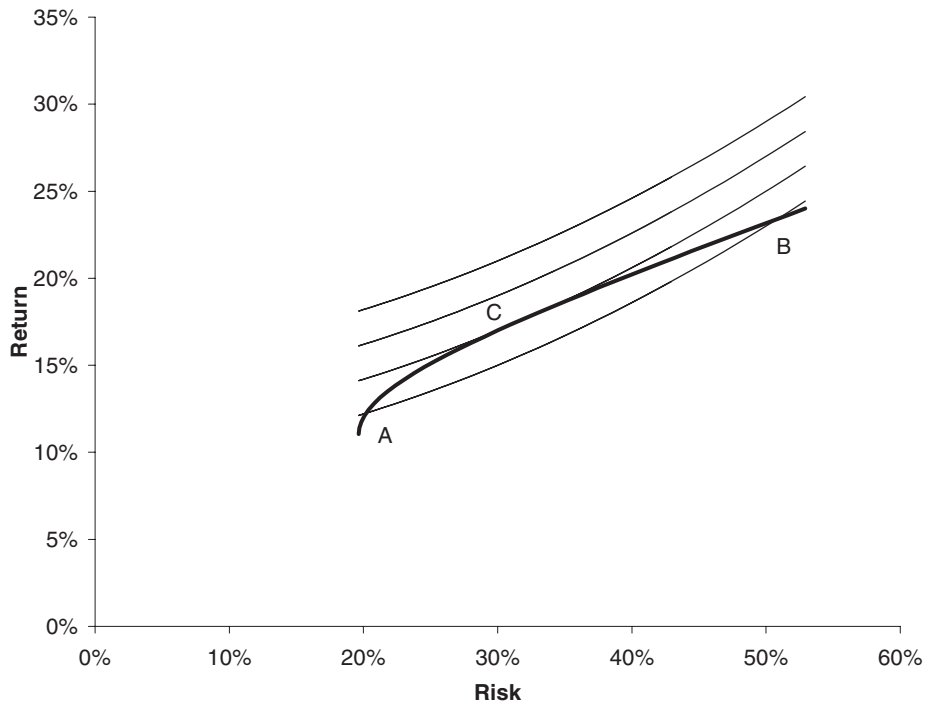


Figure 21.6 The efficient frontier and indifference curves.

efficient frontier. The curves are called by this name because they are meant to represent lines along which the investor is indifferent to the risk/reward tradeoff. An investor wants high return and low risk. Faced with portfolios A and B in the figure, he sees A with low return and low risk, but B has a better reward at the cost of greater risk. The investor is indifferent between these two. However, C is better than both of them, being on a preferred curve.



21.5 MARKOWITZ IN PRACTICE

The inputs to the Markowitz model are expected returns, volatilities and correlations. With N assets this means $N + N + N(N - 1)/2$ parameters. Most of these cannot be known accurately (do they even exist?), only the volatilities are at all reliable. Having input these parameters, we must optimize over all weights of assets in the portfolio: choose a portfolio risk and

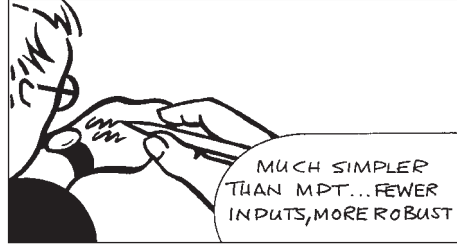
find the weights that make the return on the portfolio a maximum subject to this volatility. This is a very time-consuming process computationally unless one only has a small number of assets.

The problem with the practical implementation of this model was one of the reasons for development of the simpler model of the next section.

21.6 CAPITAL ASSET PRICING MODEL

Before discussing the **Capital Asset Pricing Model** or **CAPM** we must introduce the idea of a security's beta. The beta, β_i , of an asset relative to a portfolio M is the ratio of the covariance between the return on the security and the return on the portfolio to the variance of the portfolio. Thus

$$\beta_i = \frac{\text{Cov}[R_i R_M]}{\text{Var}[R_M]}.$$



21.6.1 The single-index model

I will now build up a **single-index model** and describe extensions later. I will relate the return on all assets to the return on a representative index, M . This index is usually taken to be a wide-ranging stock market index in the single-index model. We write the return on the i th asset as

$$R_i = \alpha_i + \beta_i R_M + \epsilon_i.$$

Using this representation we can see that the return on an asset can be decomposed into three parts: a constant drift, a random part common with the index M and a random part uncorrelated with the index, ϵ_i . The random part ϵ_i is unique to the i th asset, and has mean zero. Notice how all the assets are related to the index M but are otherwise completely uncorrelated. In Figure 21.7 is shown a plot of returns on Walt Disney stock against returns on the S&P500; α and β can be determined from a linear regression analysis. The data used in this plot ran from January 1985 until almost the end of 1997.

The expected return on the index will be denoted by μ_M and its standard deviation by σ_M . The expected return on the i th asset is then

$$\mu_i = \alpha_i + \beta_i \mu_M$$

and the standard deviation

$$\sigma_i = \sqrt{\beta_i^2 \sigma_M^2 + \sigma_{\epsilon_i}^2}$$

where σ_{ϵ_i} is the standard deviation of ϵ_i .

If we have a portfolio of such assets then the return is given by

$$\frac{\delta \Pi}{\Pi} = \sum_{i=1}^N W_i R_i = \left(\sum_{i=1}^N W_i \alpha_i \right) + R_M \left(\sum_{i=1}^N W_i \beta_i \right) + \sum_{i=1}^N W_i \epsilon_i.$$

From this it follows that

$$\mu_{\Pi} = \left(\sum_{i=1}^N W_i \alpha_i \right) + E[R_M] \left(\sum_{i=1}^N W_i \beta_i \right).$$

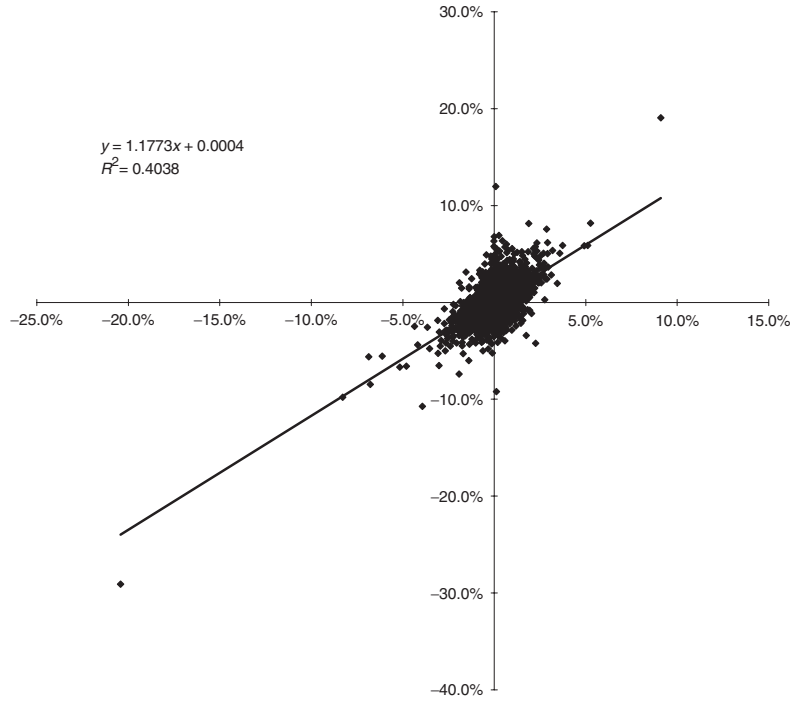


Figure 21.7 Returns on Walt Disney stock against returns on the S&P500.

Let us write

$$\alpha_{\Pi} = \sum_{i=1}^N W_i \alpha_i \quad \text{and} \quad \beta_{\Pi} = \sum_{i=1}^N W_i \beta_i,$$

so that

$$\mu_{\Pi} = \alpha_{\Pi} + \beta_{\Pi} E[R_M] = \alpha_{\Pi} + \beta_{\Pi} \mu_M.$$

Similarly the risk in Π is measured by

$$\sigma_{\Pi} = \sqrt{\sum_{i=1}^N \sum_{j=1}^N W_i W_j \beta_i \beta_j \sigma_M^2 + \sum_{i=1}^N W_i^2 e_i^2}.$$

If the weights are all about the same, N^{-1} , then the final terms inside the square root are also $O(N^{-1})$. Thus this expression is, to leading order as $N \rightarrow \infty$,

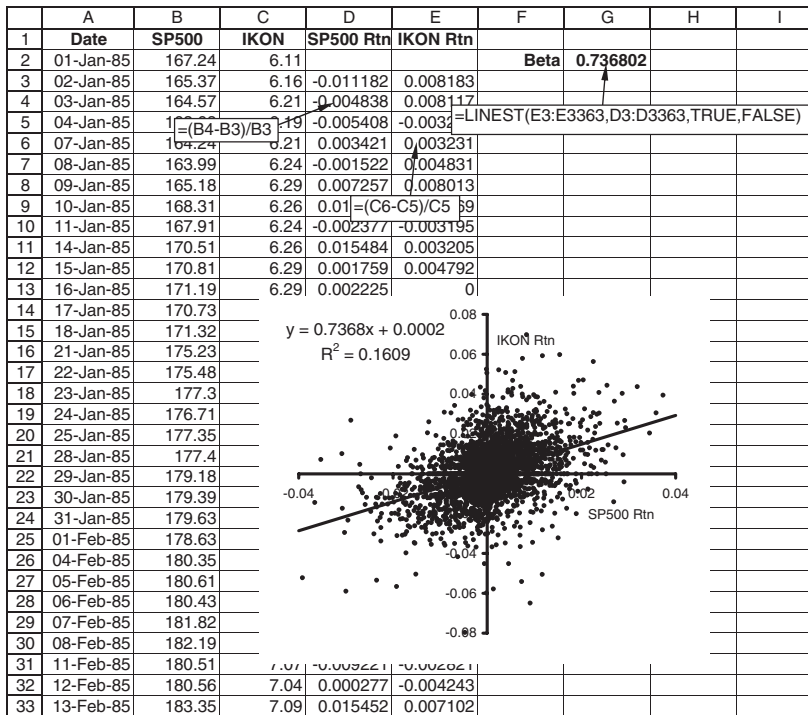
$$\sigma_{\Pi} = \left| \sum_{i=1}^N W_i \beta_i \right| \sigma_M = |\beta_{\Pi}| \sigma_M.$$

Observe that the contribution from the uncorrelated ϵ s to the portfolio vanishes as we increase the number of assets in the portfolio: the risk associated with the ϵ s is called **diversifiable risk**. The remaining risk, which is correlated with the index, is called **systematic risk**.

Time Out...

Finding beta

It's really easy to find beta from asset returns and index returns data using Excel. See the spreadsheet below. I've used the Add Trendline option when drawing the plot. This finds the best-fit straight line through the data, the slope of which is the beta.



21.6.2 Choosing the optimal portfolio

The principle is the same as the Markowitz model for optimal portfolio choice. The only difference is that there are a lot fewer parameters to be input, and the computation is a lot faster.

The procedure is as follows. Choose a value for the portfolio return μ_{Π} . Subject to this constraint, minimize σ_{Π} . Repeat this minimization for different portfolio returns to obtain efficient frontier. The position on this curve is then a subjective choice.

21.7 THE MULTI-INDEX MODEL

The model presented above is a single-index model. The idea can be extended to include further representative indices. For example, as well as an index representing the stock market one might include an index representing bond markets, an index representing currency markets or even an economic index if it is believed to be relevant in linking assets. In the multi-index model we write each asset's return as

$$R_i = \alpha_i + \sum_{j=1}^n \beta_{ji} R_j + \epsilon_i,$$

where there are n indices with return R_j . The indices can be correlated to each other. Similar results to the single-index model follow.

It is usually not worth having more than three or four indices. The fewer the parameters, the more robust will be the model. At the other extreme is the Markowitz model with one index per asset.

21.8 COINTEGRATION

Whether you use MPT or CAPM you will always worry about the accuracy of your parameters. Both of these methods are only as accurate as the input data, CAPM being more reliable than MPT generally speaking, because it has fewer parameters. There is another method which is gaining popularity, and which I will describe here briefly. It is unfortunately a complex technique requiring sophisticated statistical analysis (to do it properly) but which at its core makes a lot of sense. Instead of asking whether two series are correlated we ask whether they are **cointegrated**.

Two stocks may be perfectly correlated over short timescales yet diverge in the long run, with one growing and the other decaying. Conversely, two stocks may follow each other, never being more than a certain distance apart, but with any correlation, positive, negative or varying. If we are delta hedging then maybe the short timescale correlation matters, but not if we are holding stocks for a long time in an unhedged portfolio. To see whether two stocks stay close together we need a definition of **stationarity**. A time series is stationary if it has finite and constant mean, standard deviation and autocorrelation function. Stocks, which tend to grow, are not stationary. In a sense, stationary series do not wander too far from their mean.

We can see the difference between stationary and non-stationary with our first coin-tossing experiment. The time series given by 1 every time we throw a head and -1 every time we throw a tail is stationary. It has a mean of zero, a standard deviation of 1 and an autocorrelation function that is zero for any non-zero lag. But what if we add up the results, as we might do if we are betting on each toss? This time series is non-stationary. This is because the standard deviation of the sum grows like the square root of the number of throws. The mean may be zero but the sum is wandering further and further away from that mean.

Testing for the stationarity of a time series X_t involves a linear regression to find the coefficients a , b and c in

$$X_t = aX_{t-1} + b + ct.$$

If it is found that $|a| > 1$ then the series is unstable. If $-1 \leq a < 1$ then the series is stationary. If $a = 1$ then the series is non-stationary. As with all things statistical, we can only say that our value for a is accurate with a certain degree of confidence. To decide whether we have got a stationary or non-stationary series requires us to look at the Dickey–Fuller statistic to estimate the degree of confidence in our result. From this point on the subject of cointegration gets complicated.

How is this useful in finance? Even though individual stock prices might be non-stationary it is possible for a linear combination (i.e. a portfolio) to be stationary. Can we find λ_i , with $\sum_{i=1}^N \lambda_i = 1$, such that

$$\sum_{i=1}^N \lambda_i S_i$$

is stationary? If we can, then we say that the stocks are cointegrated.

For example, suppose we find that the S&P500 is cointegrated with a portfolio of 15 stocks. We can then use these 15 stocks to **track the index**. The error in this tracking portfolio will have constant mean and standard deviation, so should not wander too far from its average. This is clearly easier than using all 500 stocks for the tracking (when, of course, the tracking error would be zero).

We don't have to track the index, we could track anything we want, such as $e^{0.2t}$ to choose a portfolio that gets a 20% return. We could analyze the cointegration properties of two related stocks, Nike and Reebok, for example, to look for relationships. This would be pairs trading. Clearly there are similarities with MPT and CAPM in concepts such as means and standard deviations. The important difference is that cointegration assumes far fewer properties for the individual time series. Most importantly, volatility and correlation do not appear explicitly.

21.9 PERFORMANCE MEASUREMENT

If one has followed one of the asset allocation strategies outlined above, or just traded on gut instinct, can one tell how well one has done? Were the outstanding results because of an uncanny natural instinct, or were the awful results simply bad luck?

The ideal performance would be one for which returns outperformed the risk-free rate, but *in a consistent* fashion. Not only is it important to get a high return from portfolio management, but one must achieve this with as little randomness as possible.

The two commonest measures of 'return per unit risk' are the **Sharpe ratio** of 'reward to variability' and the **Treynor ratio** of 'reward to volatility'. These are defined as follows:

$$\text{Sharpe ratio} = \frac{\mu_{\Pi} - r}{\sigma_{\Pi}}$$

and

$$\text{Treynor ratio} = \frac{\mu_{\Pi} - r}{\beta_{\Pi}}.$$

In these μ_{Π} and σ_{Π} are the *realized* return and standard deviation for the portfolio over the period. The β_{Π} is a measure of the portfolio's volatility. The Sharpe ratio is usually used when the portfolio is the whole of one's investment and the Treynor ratio when one

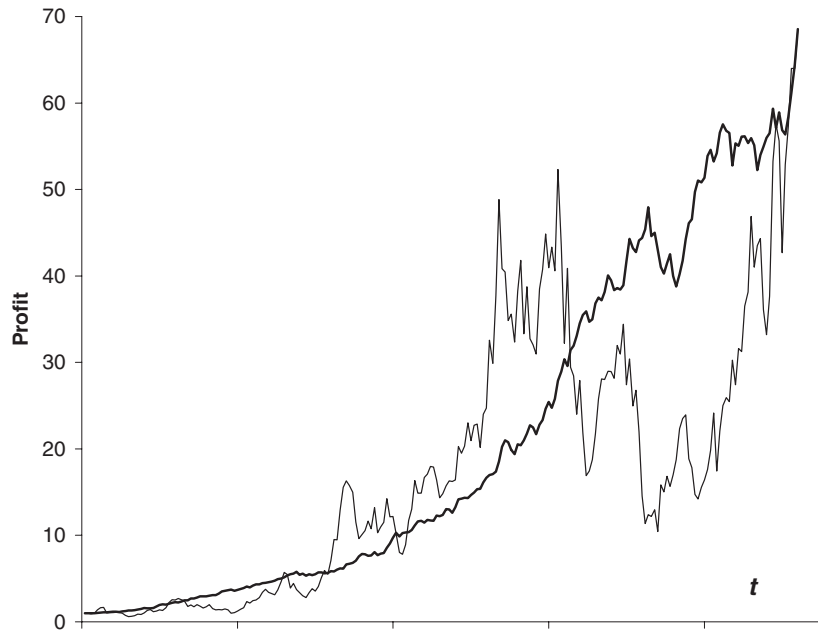


Figure 21.8 A good and a bad manager; same returns, different variability.

is examining the performance of one component of the whole firm's portfolio, say. When the portfolio under examination is highly diversified the two measures are the same (up to a factor of the market standard deviation).

In Figure 21.8 we see the portfolio value against time for a good manager and a bad manager.

21.10 SUMMARY

Portfolio management and asset allocation are about taking risks in return for a reward. The questions are how to decide how much risk to take, and how to get the best return. But derivatives theory is based on not taking any risk at all, and so I have spent little time on portfolio management in the book. On the other hand, as I have stressed, there is so much uncertainty in the subject of finance that elimination of risk is well-nigh impossible and the ideas behind portfolio management should be appreciated by anyone involved in derivatives theory or practice. I have tried to give the flavor of the subject with only the easiest-to-explain mathematics, the following sources will prove useful to anyone wanting to pursue the subject further.

FURTHER READING

- See Markowitz's original book for all the details of MPT, Markowitz (1959).
- One of the best texts on investments, including chapters on portfolio management, is Sharpe (1985).

- For a description of cointegration and other techniques in econometrics see Hamilton (1994) and Hendry (1995).
- See Farrell (1997) for further discussion of portfolio performance.
- I have not discussed the subject of continuous-time asset allocation (yet), but the elegant subject is explained nicely in the collection of Robert Merton's papers, Merton (1992).
- Transaction costs can have a big effect on portfolios that are supposed to be continuously rebalanced. See Morton & Pliska (1995) for a model with costs, and Atkinson & Wilmott (1995), Atkinson *et al.* (1997) and Atkinson & Al-Ali (1997) for asymptotic results.
- For a description of chaos-based methods in finance, and how they won the First International Nonlinear Financial Forecasting Competition, see Alexander & Giblin (1997).
- For a review of current thinking in risk management see Alexander (1998).

EXERCISES

1. Work out the efficient frontier for the following set of assets:

Asset	μ	σ
A	0.08	0.12
B	0.10	0.12
C	0.10	0.15
D	0.14	0.20

The correlation coefficients between the four assets are given by

$$\rho = \begin{pmatrix} 1 & 0.2 & 0.5 & 0.3 \\ 0.2 & 1 & 0.7 & 0.4 \\ 0.5 & 0.7 & 1 & 0.9 \\ 0.3 & 0.4 & 0.9 & 1 \end{pmatrix}.$$

2. Find the efficient frontier for the assets in the table above, when asset D is replaced by a risk-free asset, E, which has a mean of 0.10 over our time horizon of two years. Asset E is uncorrelated with the other assets.
3. Where should we be on the efficient frontier in question 1 if we wish to minimize the chance of a return less than 0.05?
4. What are the economic significances of α and β in the Capital Asset Pricing Model and how are they measured or estimated in practice?

